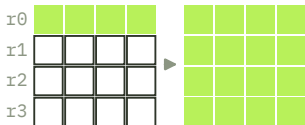


broadcast (root r0)



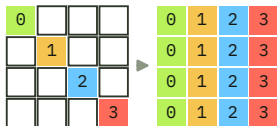
reduce (root r0)



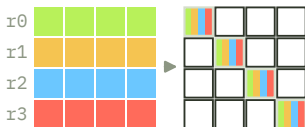
all-reduce



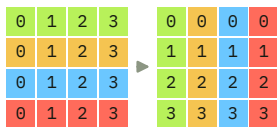
all-gather



reduce-scatter



all-to-all



colour = rank the data came from; digit = chunk index
 striped = summed over all 4 ranks
 all-to-all: chunk k of rank r goes to rank k